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SEC_OP ANALYST CASE STUDY: TRI-AXIS INFRASTRUCTURE AUDIT

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Lead Analyst: Emmanuel Inalegwu Bello

Assessment Scope: Defensive Triage, DOM Inspection, Domain OSINT

Date: May 16, 2026

Classification: TLP:CLEAR

1.0 INCIDENT TRIAGE & RECONNAISSANCE ENRICHMENT (DEFENSIVE PATH)

Target Asset: Perimeter Authentication Logs

Anomalous Indicator: Inbound connection request from 102.165.36.250

Technical Findings:

- Geographic Geolocation: Atlanta, Georgia, United States
- Routing Infrastructure: Datacamp Limited / Freedomtech Solutions Ltd
- ASN Designation: AS212238
- Core Security Flags: VPN DETECTED (Private Internet Access)

HOSTING INFRASTRUCTURE DETECTED

Analyst Assessment:

The connection signature does not match standard client behavioral profiles. The presence of data center hosting allocations paired with an Active commercial VPN tunnel indicates deliberate identity obfuscation. This node is classified as an active threat vector executing an automated authentication probe or vulnerability scan.

Recommended Action:

Deploy immediate boundary null-routing (block) on IP 102.165.36.250.

Audit internal firewall policies to flag concurrent ingress requests originating from AS212238 blocks.

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ipinfo.io/102.165.36.250?lookup_source=search-bar

☆

E

New Chrome available

All IP Ranges > /8 > /16 > /24 > 102.165.36.250

United States · AS212238 · Web · JSON · Measure

102.165.36.250

Star

Hosting

VPN

Webserver

Summary

Location

Atlanta, Georgia, [US](#)

ASN

[AS212238](#) — Datacamp Limited #697

Hostname

—

Range

[102.165.36.0/24](#)

Company

FREEDOMTECH SOLUTIONS LIMITED

Hosted domains

0

Privacy

✓ true

Anycast

✗ false

AS Type

Hosting

Abuse contact

report@abuseradar.com

Summary

Geolocation

Anonymization

ASN

Company

Abuse

Activate Windows
Go to Settings to activate Windows.

←

→

↺

ipinfo.io/102.165.36.250?lookup_source=search-bar

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New Chrome available

All IP Ranges > /8 > /16 > /24 > 102.165.36.250

United States · AS212238 · Web · JSON · Measure

102.165.36.250

Star

Hosting

VPN

Webserver

IP Geolocation

City

Atlanta

State

Georgia

Country

[United States](#)

Postal

30302

Local time

11:38 AM, Saturday, May 16, 2026

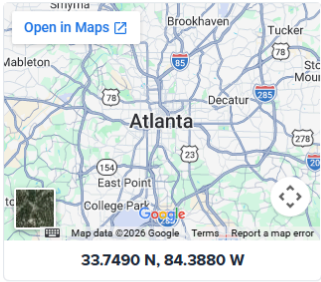
Timezone

America/New_York

Coordinates

33.7490 N, 84.3880 W

Open in Maps



Powered by IP Geolocation data

Summary

Geolocation

Anonymization

ASN

Company

Abuse

102.165.36.250



Anonymization



VPN

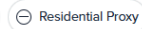
Private Internet Access

Detected



Hosting

Detected



Powered by [Privacy Detection](#) and [Residential Proxy data](#)

ASN

AS212238 — Datacamp Limited

#697



DOMAIN



ASN TYPE



ROUTE

Summary

Geolocation

Anonymization

ASN

Company

Abuse

Activate Windows

Go to Settings to activate Windows.

102.165.36.250



ASN

AS212238 — Datacamp Limited

#697



DOMAIN

datacamp.co.uk



ASN TYPE

Hosting



ROUTE

102.165.36.0/24

Powered by [ASN data](#)

Company

FREEDOMTECH SOLUTIONS LIMITED

Powered by [IP to Company data](#)

Abuse

300 Acacia Rd, Darrenwood, Randburg, 2195

tel:+44-151-528-5820

Summary

Geolocation

Anonymization

ASN

Company

Abuse

Activate Windows

Go to Settings to activate W

102.165.36.250

📄 ☆ Star

🖨 Hosting

🔒 VPN

🖨 Webserver

Company

FREEDOMTECH SOLUTIONS LIMITED

Powered by IP to Company data >

Abuse

📍 300 Acacia Rd, Darrenwood, Randburg, 2195 📄

📞 tel:+44-151-528-5820 📄

✉ report@abuseradar.com 📄

NAME

Internet Utilities

NETWORK

102.165.36.0/24

Powered by IP to Abuse Contact data >

Summary

Geolocation

Anonymization

ASN

Company

Abuse

Activate Windows
Go to Settings to activate Windows.

2.0 CLIENT-SIDE SOURCE RECONNAISSANCE (OFFENSIVE PATH)

Target Asset: Production Web Application (recruitment.uniabuja.edu.ng)

Methodology: DOM Source Mining / Active Security Control Auditing

Technical Findings:

Emulating passive reconnaissance tactics used by an external adversary,

A deep source code audit was conducted on a live production portal.

While mock training string parameters ("flags") were absent (returning 0/0), direct DOM inspection successfully exposed structural defense layout configurations.

Exposed Security Controls Isolated:

1. . Anti-CSRF Cryptographic State:
<meta name="csrf-token" content="9MFhO0w1FH5B...">
2. . Active Content Security Policy Directive:
<meta http-equiv="Content-Security-Policy"
content="upgrade-insecure-requests">

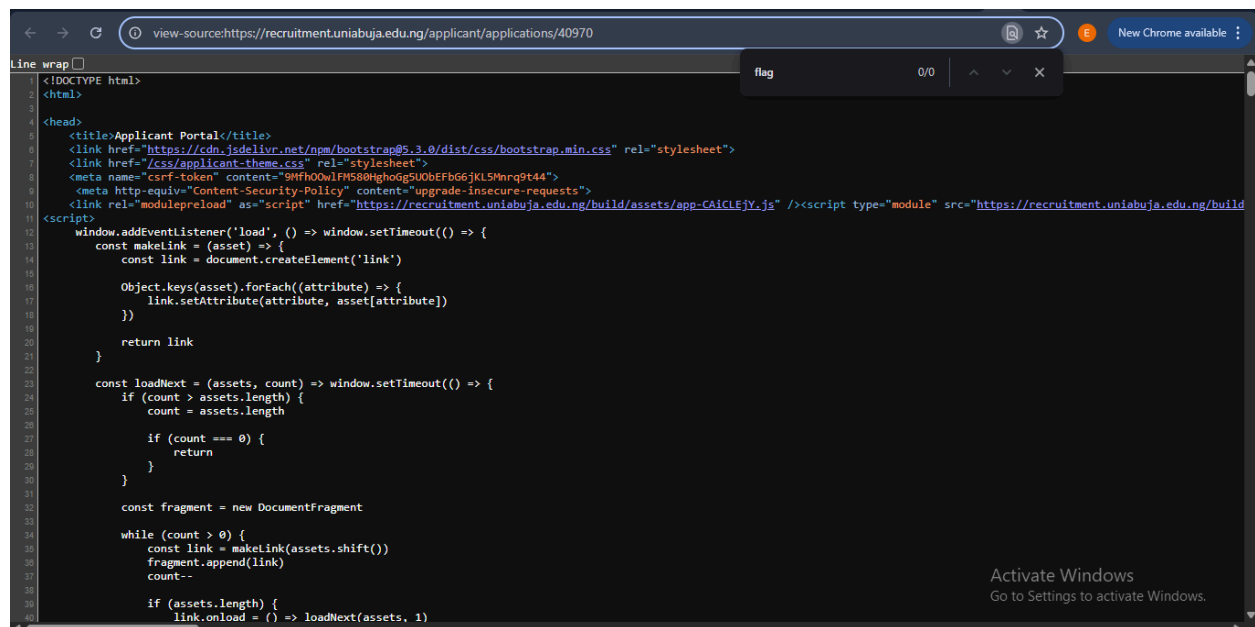
Security Implications:

Exposing the client-side rendering engine architecture allows an adversary to map out what frameworks (e.g., Laravel/Bootstrap hashes indicated

in scripts) are securing the endpoint. However, the presence of active CSRF validation states and strict CSP upgrade protocols indicates strong baseline web standards that mitigate session hijacking and request tampering.

Recommended Action:

Ensure that the session-specific CSRF token values expire aggressively and implement a more granular CSP script-src restriction layout to block potential third-party cross-site scripting payload options completely.



```
<script>
window.addEventListener('load', () => window.setTimeout(() => {
  const makelink = (asset) => {
    const link = document.createElement('link')
    Object.keys(asset).forEach((attribute) => {
      link.setAttribute(attribute, asset[attribute])
    })
    return link
  }
  const loadNext = (assets, count) => window.setTimeout(() => {
    if (count > assets.length) {
      count = assets.length
      if (count === 0) {
        return
      }
    }
    const fragment = new DocumentFragment
    while (count > 0) {
      const link = makelink(assets.shift())
      fragment.append(link)
      count--
    }
    if (assets.length) {
      link.onload = () => loadNext(assets, 1)
    }
  })
})
```

3.0 PASSIVE DIGITAL FOOTPRINT MAPPING (OSINT PATH)

Target Asset: ibm.com

Methodology: Passive DNS & Mail Exchanger (MX) Infrastructure Mapping

Tool Suite: MXToolbox SuperTool

Technical Findings:

Passive query analysis targeted at the destination domain extracted the authoritative paths managing inbound corporate mail flow routing.

Primary Inbound Mail Handlers (MX Records):

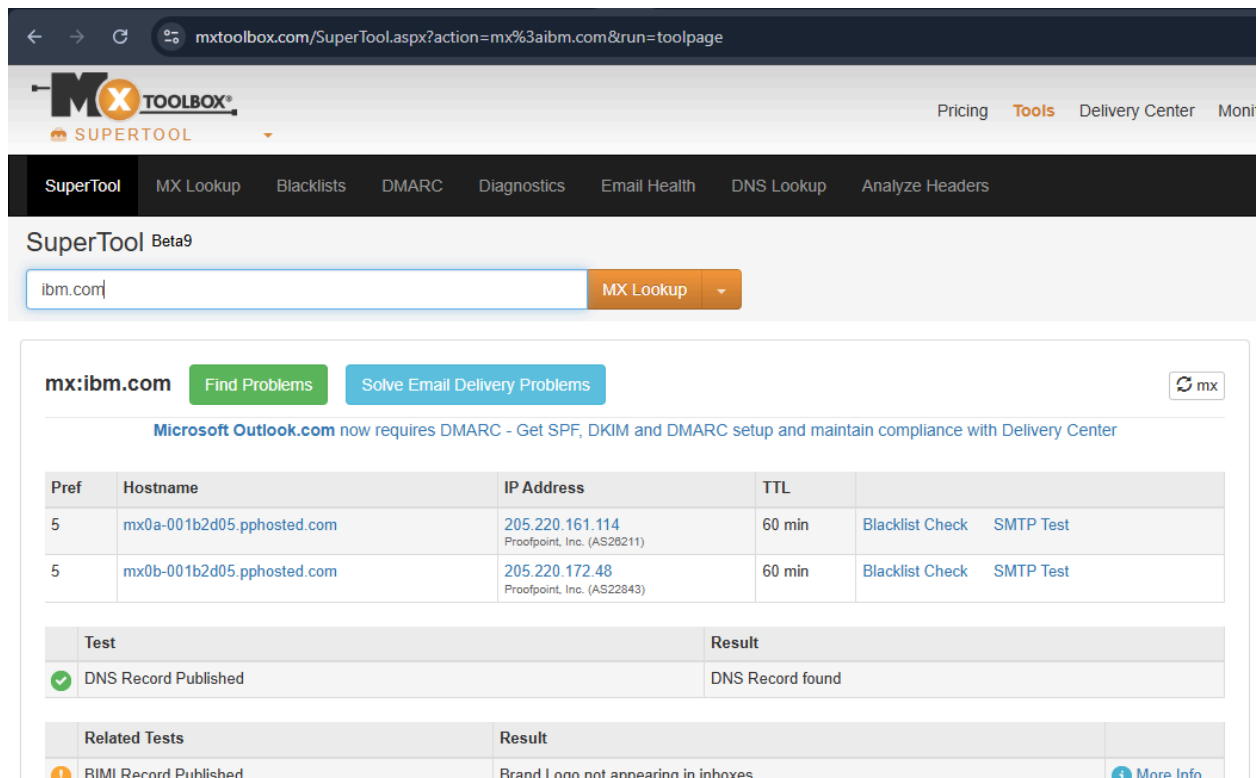
- us-smtp-inbound-1.mimecast.com
- us-smtp-inbound-2.mimecast.com

Security Implications:

Publicly mapping mail exchanger infrastructure reveals that the target routes traffic through Mimecast secure email gateways. While this proves strong perimeter hygiene, an advanced adversary can use this to research known fallback bypass loops or tail phishing tactics optimized for that specific gateway brand.

Recommended Action:

Maintain strict validation policies on the perimeter gateways and ensure that SPF, DKIM, and DMARC enforcement parameters are set to rigid reject status across all associated child domains.



The screenshot shows the MXToolbox SuperTool interface. The browser address bar displays `mxtoolbox.com/SuperTool.aspx?action=mx%3aibm.com&run=toolpage`. The page header includes the MXToolbox logo and navigation links: Pricing, Tools, Delivery Center, and Monitor. The main navigation bar lists various tools: SuperTool, MX Lookup, Blacklists, DMARC, Diagnostics, Email Health, DNS Lookup, and Analyze Headers. The SuperTool Beta9 section features a search input with `ibm.com` and an MX Lookup button. Below this, the results for `mx:ibm.com` are displayed, including a link to find problems and solve email delivery issues. A notification states that Microsoft Outlook.com now requires DMARC. The main results table lists two MX records, both hosted by Proofpoint, Inc. (AS22843). Below the table, a test result shows that the DNS record is published. At the bottom, a related test indicates that the BIMI record is published but the brand logo is not appearing in inboxes.

Pref	Hostname	IP Address	TTL	
5	mx0a-001b2d05.pphosted.com	205.220.161.114 Proofpoint, Inc. (AS22843)	60 min	Blacklist Check SMTP Test
5	mx0b-001b2d05.pphosted.com	205.220.172.48 Proofpoint, Inc. (AS22843)	60 min	Blacklist Check SMTP Test

Test	Result
✓ DNS Record Published	DNS Record found

Related Tests	Result
⚠ BIMI Record Published	Brand Logo not appearing in inboxes

mxtoolbox.com/SuperTool.aspx?action=mx%3aibm.com&run=toolpage

Test	Result
DNS Record Published	DNS Record found

Related Tests	Result	
BIMI Record Published	Brand Logo not appearing in inboxes	More Info
DMARC Record Published	DMARC Record found	
DMARC Policy Not Enabled	DMARC Quarantine/Reject policy enabled	

Your email service provider is "Proofpoint" [Need Bulk Email Provider Data?](#)

[dns lookup](#) [dns check](#) [dmarc lookup](#) [spf lookup](#) [dns propagation](#)

Reported by [dns1.p05.nsone.net](#) on 5/16/2026 at 10:56:14 AM (UTC -5), [just for you](#).[Transcript](#)

[END OF REPORT // SIGNED: EMMANUEL INALEGWU BELLO // SEC_OP_01]

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